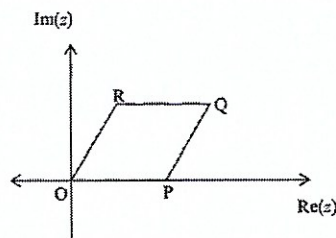


Complex Number 16

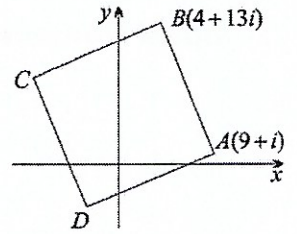
1. If $|z_1 + z_2| = |z_1 - z_2|$, find the possible values of $\arg\left(\frac{z_1}{z_2}\right)$.
2. If $z_1 = 24 + 7i$ and $|z_2| = 6$, find the greatest and least values of $|z_1 + z_2|$.
3. Express $1 + i$ and $1 - i$ in modulus/argument form. Use De Moivre's theorem to evaluate $(1 + i)^{20} + (1 - i)^{20}$.
4. Express $-1 + \sqrt{3}i$ in modulus/argument form. Use De Moivre's theorem to show that $(-1 + \sqrt{3}i)^n + (-1 - \sqrt{3}i)^n = 2^{n+1} \cos\left(\frac{2n\pi}{3}\right)$, n a positive integer. Evaluate this expression in each of the cases $n = 3m, n = 3m \pm 1$ where m is a positive integer.
5. Use De Moivre's theorem to solve the equation $z^5 = 1$. Show that the points representing the five roots of this equation on an Argand diagram form the vertices of a regular pentagon of area $\frac{5}{2} \sin \frac{2\pi}{5}$ and perimeter $10 \sin \frac{\pi}{5}$.
6. Use De Moivre's theorem to solve $z^5 = -1$. By grouping the complex roots in conjugate pairs, show that $z^5 + 1 = (z + 1)\left(z^2 - 2z \cos \frac{\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{3\pi}{5} + 1\right)$.
7. a. If ω is a complex solution of the equation $z^3 = 1$ show that ω^2 is also a solution of the equation and show that: $1 + \omega + \omega^2 = 0$.
b. Given that ω is also a root of $z^3 + bz^2 + cz - 12 = 0$, find the values of b and c given they are both real.
8. a. Solve the equation $z^5 - 1 = 0$ over the complex number field.
b. Hence factorise $z^5 - 1$ in terms of real linear and quadratic factors.
c. Hence, show that $\cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4}$.
9. If $z = r(\cos \theta + i \sin \theta)$ show that $\frac{z}{z^2 + r^2}$ is purely real and find its value.
10. If $1, w, w^2$ are the 3 cube roots of unity and given that $1 + w + w^2 = 0$:
a. Show that $1 + w$ is a root of the equation $z^3 - 3z^2 + 3z - 2 = 0$.
b. Find the integer root of this equation AND also the third roots in terms of w .
11. $OPQR$ is a rhombus. O lies at the origin, P on the real axis and R corresponds to the complex number $1 + \sqrt{3}i$.
a. Find the complex number corresponding to Q .
b. If the figure is rotated anticlockwise by 60° about O to form a new rhombus $OP'Q'R'$, show this on an Argand Diagram and find the complex number corresponding to the vertex at Q' .



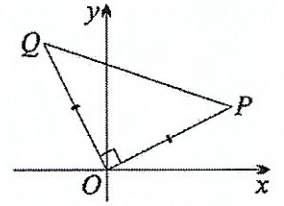
12. a. Find the locus of $\operatorname{Re}(z - 2) + 2 = |z - 2|$
b. Find the range of $\arg(z)$.
13. a. Given $|z| = 1$, show that $z^n + z^{-n} = 2 \cos n\theta$.
b. Solve $z^6 + 1 = 0$ and hence factorise $z^6 + 1$ into real quadratic factors.
c. Hence deduce that $\cos 3\theta = \cos \theta (2 \cos \theta + \sqrt{3})(2 \cos \theta - \sqrt{3})$

14. The diagram on the right shows a square $ABCD$ in the complex plane. The vertices A and B represent the complex numbers $9 + i$ and $4 + 13i$ respectively. Find the complex numbers that correspond to:

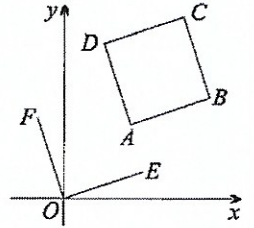
- The vector $\overrightarrow{AB}$
- The vertex D



15. In the diagram, the point P and Q correspond to the complex numbers z and w respectively. The triangle OPQ is isosceles and the angle POQ is a right angle. Prove that $z^2 + w^2 = 0$.

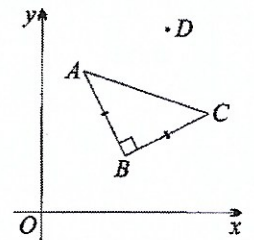


16. In this Argand diagram, $ABCD$ is a square, and OE and OF are parallel and equal in length to AB and AD respectively. The vertices A and B correspond to the complex number w_1 and w_2 respectively. What complex number correspond to the points E, F, C and D ?

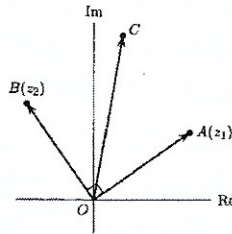


17. In this diagram, the vertices of a triangle ABC are represented by the complex numbers z_1, z_2 and z_3 respectively. The triangle is isosceles, and right-angled at B .

- Explain why $(z_1 - z_2)^2 + (z_3 - z_2)^2 = 0$
- Suppose that D is the point such that $ABCD$ is a square. Find in terms of z_1, z_2 and z_3 , the complex number that the point D represents.



18. In the Argand diagram below $z_1 = \cos \theta + i \sin \theta$. The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the complex numbers z_1 and z_2 . $OACB$ forms a square with $\overrightarrow{OC}$ being a diagonal.



- Show that $z_1 + z_2 = (1 + i)z_1$
- Show that $(z_1 + z_2)(\overline{z_1 - z_2}) = 2i$

Answer

- $\pm \frac{\pi}{2}$
- 31, 19
- $\sqrt{2} \operatorname{cis} \left(\pm \frac{\pi}{4} \right), -2048$
- $2 \operatorname{cis} \left(\frac{2\pi}{3} \right); 2^{n+1}, -2^n$
- $z = 1, \operatorname{cis} \left(\pm \frac{2\pi}{5} \right), \operatorname{cis} \left(\pm \frac{4\pi}{5} \right)$
- $z = -1, \operatorname{cis} \left(\pm \frac{\pi}{5} \right), \operatorname{cis} \left(\pm \frac{3\pi}{5} \right)$
- $b = -11, c = -11$
- $z = \operatorname{cis} \frac{2k\pi}{5}, k = 0, 1, 2, 3, 4$
 - $z^5 - 1 = (z - 1) \left(z^2 - 2 \cos \frac{2\pi}{5} z + 1 \right) \left(z^2 - 2 \cos \frac{4\pi}{5} z + 1 \right)$
- $\frac{1}{2r \cos \theta}$
- 2 is an integer root, $-w$ is the third root.
 - $Q = 3 + \sqrt{3}i$ $Q = 2\sqrt{3}i$
- $y^2 = 4(x - 1)$ $-\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$
 - $-5 + 12i$ $-3 - 4i$
- E represents $w_2 - w_1$, F represents $i(w_2 - w_1)$, C represents $w_2 + i(w_2 - w_1)$ and D represents $w_1 + i(w_2 - w_1)$
- Vectors $\overrightarrow{BA}$ and $\overrightarrow{BC}$ represents $z_1 - z_2$ and $z_3 - z_2$ respectively, and $\overrightarrow{BA}$ is the anticlockwise rotation of $\overrightarrow{BC}$ through 90° about B . So $z_1 - z_2 = i(z_3 - z_2)$. Squaring both sides gives the result. $z_1 - z_2 + z_3$